emotion.py - C:\Users\jaswa\OneDrive\文档\emotion.py (3.13.14)

File Edit Format Run Options Window Help

```python
from collections import deque

def solve_8_puzzle(start, goal):
    queue = deque([(start, [])])
    visited = {tuple(start)}

    while queue:
        state, path = queue.popleft()

        if state == goal:
            return path + [state]

        zero = state.index(0)
        row, col = divmod(zero, 3)

        moves = [
            (-1, 0),  # Up
            (1, 0),   # Down
            (0, -1),  # Left
            (0, 1)    # Right
        ]

        for dr, dc in moves:
            nr, nc = row + dr, col + dc

            if 0 <= nr < 3 and 0 <= nc < 3:
                new_zero = nr * 3 + nc
                new_state = state.copy()

                new_state[zero], new_state[new_zero] = \
                    new_state[new_zero], new_state[zero]

                if tuple(new_state) not in visited:
                    visited.add(tuple(new_state))
                    queue.append((new_state, path + [state]))

    return None


start = [1, 2, 3,
         4, 0, 6,
         7, 5, 8]

goal = [1, 2, 3,
        4, 5, 6,
        7, 8, 0]
```

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:\Users\jaswa\OneDrive\文档\emotion.py ==============
==
Solution found:
[1, 2, 3]
[4, 0, 6]
[7, 5, 8]

[1, 2, 3]
[4, 5, 6]
[7, 0, 8]

[1, 2, 3]
[4, 5, 6]
[7, 8, 0]

>>>
```

Ln: 18 Col: 0

Ln: 18 Col: 28

31°C Light rain

Search

ENG IN

19:26 02-10-2026

emotion.py - C:\Users\jaswa\OneDrive\文档\emotion.py (3.13.14)

File Edit Format Run Options Window Help

```
N = 8

board = [-1] * N


def is_safe(row, col):
    for i in range(row):
        # Same column
        if board[i] == col:
            return False

        # Same diagonal
        if abs(board[i] - col) == abs(i - row):
            return False

    return True


def solve(row):
    if row == N:
        return True

    for col in range(N):
        if is_safe(row, col):
            board[row] = col

            if solve(row + 1):
                return True

            board[row] = -1

    return False


if solve(0):
    print("Solution:")
    for row in range(N):
        for col in range(N):
            if board[row] == col:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()
else:
    print("No solution exists.")
```

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:\Users\jaswa\OneDrive\文档\emotion.py ==============
==
Solution:
Q . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
>>>
```

Ln: 14 Col: 0

Ln: 35 Col: 12

30°C Partly cloudy

Search

ENG IN

19:29 02-10-2026

emotion.py - C:\Users\jaswa\OneDrive\文档\emotion.py (3.13.14)

File Edit Format Run Options Window Help

```python
from collections import deque


def water_jug():
    capacity1 = 4
    capacity2 = 3
    target = 2

    start = (0, 0)

    queue = deque([(start, [])])
    visited = {start}

    while queue:
        (x, y), path = queue.popleft()

        if x == target or y == target:
            print("Solution:")
            for step in path:
                print(step)
            print("Final State:", (x, y))
            return

        states = [
            ((capacity1, y), "Fill 4L jug"),
            ((x, capacity2), "Fill 3L jug"),
            ((0, y), "Empty 4L jug"),
            ((x, 0), "Empty 3L jug")
        ]

        # Pour 4L jug into 3L jug
        amount = min(x, capacity2 - y)
        states.append(
            ((x - amount, y + amount),
             "Pour 4L jug into 3L jug")
        )

        # Pour 3L jug into 4L jug
        amount = min(y, capacity1 - x)
        states.append(
            ((x + amount, y - amount),
             "Pour 3L jug into 4L jug")
        )

        for new_state, operation in states:
            if new_state not in visited:
                visited.add(new state)
```

Ln: 55 Col: 11

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:\Users\jaswa\OneDrive\文档\emotion.py ==============
==
Solution:
Fill 3L jug
Pour 3L jug into 4L jug
Fill 3L jug
Pour 3L jug into 4L jug
Final State: (4, 2)
>>>
```

Ln: 11 Col: 0

30°C Partly cloudy

Search

ENG IN

19:30 02-10-2026

emotion.py - C:\Users\jaswa\OneDrive\文档\emotion.py (3.13.14)

File Edit Format Run Options Window Help

```
from itertools import permutations

for p in permutations(range(10), 8):

    S, E, N, D, M, O, R, Y = p

    if S == 0 or M == 0:
        continue

    SEND = 1000*S + 100*E + 10*N + D
    MORE = 1000*M + 100*O + 10*R + E
    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y

    if SEND + MORE == MONEY:
        print("SEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)

        print("S =", S)
        print("E =", E)
        print("N =", N)
        print("D =", D)
        print("M =", M)
        print("O =", O)
        print("R =", R)
        print("Y =", Y)

        break
```

Ln: 28 Col: 13

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:\Users\jaswa\OneDrive\文档\emotion.py ==============
==
SEND = 9567
MORE = 1085
MONEY = 10652
S = 9
E = 5
N = 6
D = 7
M = 1
O = 0
R = 8
Y = 2
>>>
```

Ln: 16 Col: 0

30°C
Partly cloudy

Search

ENG
IN

21:59
05-10-2026

emotion.py - C:\Users\jaswa\OneDrive\文档\emotion.py (3.13.14)

File Edit Format Run Options Window Help

```
    if right_m > 0 and right_c > right_m:
        return False

    return True

def neighbors(state):
    m, c, boat = state
    result = []

    for dm, dc in moves:

        if boat == 0:
            new_state = (m-dm, c-dc, 1)
        else:
            new_state = (m+dm, c+dc, 0)

        if valid(new_state[0], new_state[1]):
            result.append(new_state)

    return result

queue = deque([start])
parent = {start: None}

while queue:
    state = queue.popleft()

    if state == goal:
        break

    for new_state in neighbors(state):
        if new_state not in parent:
            parent[new_state] = state
            queue.append(new_state)

path = []

state = goal

while state is not None:
    path.append(state)
    state = parent[state]

path.reverse()

for state in path:
    print(state)
```

Ln: 70 Col: 16

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>

================ RESTART: C:\Users\jaswa\OneDrive\文档\emotion.py ==============
==
(3, 3, 0)
(3, 1, 1)
(3, 2, 0)
(3, 0, 1)
(3, 1, 0)
(1, 1, 1)
(2, 2, 0)
(0, 2, 1)
(0, 3, 0)
(0, 1, 1)
(1, 1, 0)
(0, 0, 1)
>>>
```

Ln: 17 Col: 0

30°C Partly cloudy

Search

ENG IN

22:00 05-10-2026

emotion.py - C:\Users\jaswa\OneDrive\文档\emotion.py (3.13.14)

File Edit Format Run Options Window Help

```
rooms = {
    "A": "Dirty",
    "B": "Dirty"
}

position = "A"

print("Initial State:", rooms)

if rooms["A"] == "Dirty":
    print("Cleaning Room A")
    rooms["A"] = "Clean"

position = "B"

if rooms["B"] == "Dirty":
    print("Cleaning Room B")
    rooms["B"] = "Clean"

print("Final State:", rooms)
```

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>

================ RESTART: C:\Users\jaswa\OneDrive\文档\emotion.py ==============
==
Initial State: {'A': 'Dirty', 'B': 'Dirty'}
Cleaning Room A
Cleaning Room B
Final State: {'A': 'Clean', 'B': 'Clean'}
>>>
```

Ln: 9 Col: 0

Ln: 21 Col: 0

30°C
Partly cloudy

Search

ENG
IN

22:01
05-10-2026

emotion.py - C:\Users\jaswa\OneDrive\文档\emotion.py (3.13.1

File Edit Format Run Options Window Help

```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

def bfs(graph, start):

    visited = set()
    queue = deque([start])

    visited.add(start)

    while queue:

        node = queue.popleft()
        print(node, end=" ")

        for neighbour in graph[node]:

            if neighbour not in visited:
                visited.add(neighbour)
                queue.append(neighbour)

bfs(graph, 'A')
```

Ln: 30 Col: 15

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>

================ RESTART: C:\Users\jaswa\OneDrive\文档\emotion.py ==============
==
A B C D E F
>>>
```

Ln: 6 Col: 0

30°C
Partly cloudy

Search

ENG
IN

22:02
05-10-2026

emotion.py - C:\Users\jaswa\OneDrive\文档\emotion.py (3.13.14)

File Edit Format Run Options Window Help

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

visited = set()

def dfs(node):

    if node not in visited:

        print(node, end=" ")

        visited.add(node)

        for neighbour in graph[node]:
            dfs(neighbour)

dfs('A')
```

Ln: 23 Col: 8

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>

================ RESTART: C:\Users\jaswa\OneDrive\文档\emotion.py ==============
==
A B D E F C
>>>
```

Ln: 6 Col: 0

30°C
Partly cloudy

Search

ENG
IN

22:03
05-10-2026

emotion.py - C:\Users\jaswa\OneDrive\文档\emotion.py (3.13.14)

File Edit Format Run Options Window Help

```
from itertools import permutations

distance = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

cities = [1, 2, 3]

minimum_cost = float('inf')
best_route = None

for route in permutations(cities):

    current_route = (0,) + route + (0,)

    cost = 0

    for i in range(len(current_route) - 1):
        cost += distance[current_route[i]][current_route[i+1]]

    if cost < minimum_cost:
        minimum_cost = cost
        best_route = current_route

print("Best Route:")

for city in best_route:
    print(city + 1, end=" ")

print("\nMinimum Cost =", minimum_cost)
```

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:\Users\jaswa\OneDrive\文档\emotion.py ==============
==
Best Route:
1 2 4 3 1
Minimum Cost = 80
>>>
```

Ln: 8 Col: 0

Ln: 33 Col: 39

30°C Partly cloudy

Search

ENG IN

22:04 05-10-2026

emotion.py - C:\Users\jaswa\OneDrive\文档\emotion.py (3.13.14)

File Edit Format Run Options Window Help

```
}

heuristic = {
    'A': 5,
    'B': 4,
    'C': 2,
    'D': 3,
    'E': 1,
    'F': 0
}

def astar(start, goal):

    queue = [(heuristic[start], 0, start, [start])]

    visited = {}

    while queue:

        f, cost, node, path = heapq.heappop(queue)

        if node == goal:
            return path, cost

        if node in visited and visited[node] <= cost:
            continue

        visited[node] = cost

        for neighbour, weight in graph[node].items():

            new_cost = cost + weight

            new_f = new_cost + heuristic[neighbour]

            heapq.heappush(
                queue,
                (new_f,
                 new_cost,
                 neighbour,
                 path + [neighbour])
            )

path, cost = astar('A', 'F')

print("Shortest Path:", path)
print("Cost:", cost)
```

Ln: 56 Col: 20

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:\Users\jaswa\OneDrive\文档\emotion.py ==============
==
Shortest Path: ['A', 'B', 'D', 'F']
Cost: 6
>>>
```

Ln: 7 Col: 0

30°C Partly cloudy

Search

ENG IN

22:05 05-10-2026